# Prime Dissections and the Optimality of Archimedes' Stomachion

**Abstract**

We introduce the notion of a *prime* (irreducible) dissection—one admitting no fault lines—and show that the solution space of any lattice dissection decomposes as a coproduct of products over independent sub-regions. Every dissection factors uniquely into prime sub-dissections, in direct analogy with the fundamental theorem of arithmetic. We present computational evidence, based on a DLX exact-cover solver applied to $12 \times 12$ lattice dissections, that Archimedes' Stomachion (14 pieces, 2144 geometric tilings) is maximal among prime dissections of comparable size. Furthermore, the distribution of tiling counts for random lattice dissections follows a Poisson-like law in the number of independent exchange moves: for a random $n$-piece dissection, the number of such moves is approximately Poisson($\lambda n$) for a small constant $\lambda \approx 0.15$, and the geometric tiling count concentrates on powers of two with an exponentially heavy right tail.

## 1 Introduction

The *Stomachion* (also *Loculus Archimedius*) is a dissection puzzle attributed to Archimedes, in which a $12 \times 12$ square is divided into 14 pieces—11 triangles, 2 quadrilaterals, and a pentagon—all with vertices at integer lattice points. The puzzle asks: in how many ways can these pieces be rearranged to tile the original square?

The answer, established computationally by Cutler in 2003 [3] shortly after the decipherment of the Archimedes Palimpsest by Netz and Noel [5], is **536**. This count identifies two tilings if one can be obtained from the other by a symmetry of the square (the dihedral group $D_4$) or by swapping congruent piece pairs. Without the piece-swap identification—counting tilings up to $D_4$ symmetry alone, so that each tiling is a geometrically distinct partition of the square into labeled regions—the count rises to **2144**.

These numbers are remarkably large. A generic 14-piece dissection of a square is *rigid*: it admits exactly one tiling. Yet the Stomachion admits over two thousand. What is special about it?

One answer is immediate: a dissection with a *fault line*—a straight line from boundary to boundary that no piece crosses—decomposes into independent sub-puzzles, and the total solution count is a product. By introducing a single horizontal fault line, one can construct 14-piece dissections with over 4600 geometric tilings. But such dissections feel like cheating: they are really two 7-piece puzzles glued together.

This leads us to the central definition of the paper.

**Definition 1.1** (Prime dissection)**.** A dissection of a convex region is *prime* (or *irreducible*) if it admits no fault line. Equivalently, no straight line from boundary to boundary can be drawn without crossing the interior of at least one piece.

The Stomachion is prime. And among prime dissections, its tiling count appears to be maximal—or at least extreme. The purpose of this paper is to make this observation precise, to develop the algebraic framework of prime factorization for dissections, and to present computational evidence for a distributional law governing tiling counts.

**Summary of results.**

(i) Every lattice dissection factors uniquely into prime sub-dissections separated by fault lines, and the solution space decomposes as a coproduct of products (Theorem 2.2).

(ii) We verify computationally that the Stomachion has 17,152 raw tilings, giving 2144 geometric tilings modulo $D_4$ and 536 modulo $D_4$ plus piece-swap symmetries.

(iii) Random $n$-piece convex lattice dissections are generically rigid or near-rigid; dissections with many tilings are exponentially rare (Section 4.2).

(iv) Reducible dissections easily surpass the Stomachion's count, but among prime dissections the Stomachion appears maximal at $n = 14$ (Conjecture 4.3).

(v) The number of independent exchange moves in a random dissection is approximately Poisson-distributed, giving rise to a power-of-two concentration in tiling counts (Conjecture 5.2).

# 2 The Coproduct Decomposition

We work throughout with *lattice dissections*: partitions of a square $[0, N]^2$ into polygonal pieces whose vertices lie on $\mathbb{Z}^2 \cap [0, N]^2$. A *tiling* of the square by a set of pieces $\mathcal{P}$ is a placement of every piece (possibly rotated or reflected) such that the pieces cover the square without overlap.

**Definition 2.1** (Fault line)**.** Let $\mathcal{D}$ be a dissection of the square $S = [0, N]^2$ into pieces $P_1, \ldots, P_n$. A *fault line* of $\mathcal{D}$ is a line segment $\ell$ connecting two points on $\partial S$ such that $\ell \cap \mathrm{int}(P_i) = \emptyset$ for every $i$. Equivalently, $\ell$ runs entirely along piece boundaries.

A fault line $\ell$ divides $S$ into two sub-regions $R^+$ and $R^-$ (or more, if $\ell$ is not a single segment). The pieces of $\mathcal{D}$ partition into those lying entirely in $R^+$ and those in $R^-$, and any tiling of $S$ restricts to independent tilings of each sub-region.

**Theorem 2.2** (Coproduct decomposition)**.** *Let $\mathcal{D}$ be a dissection of the square $S$ with fault lines $\ell_1, \ldots, \ell_k$ that divide $S$ into sub-regions $R_1, \ldots, R_m$. Let $\mathcal{P}_j \subset \mathcal{P}$ be the pieces assigned to region $R_j$ in the canonical dissection. Then*

$$\mathrm{Sol}(S, \mathcal{P}) \;=\; \coprod_{\alpha \in \mathcal{A}} \prod_{j=1}^{m} \mathrm{Sol}\big(R_j,\, \mathcal{P}_j^{\alpha}\big),$$

*where $\mathcal{A}$ indexes the valid assignments of pieces to regions (since congruent pieces might be assigned to different regions in different tilings), and each* $\mathrm{Sol}(R_j, \mathcal{P}_j^\alpha)$ *denotes the set of tilings of region $R_j$ using piece set $\mathcal{P}_j^\alpha$.*

*Proof.* Since every piece lies entirely within one sub-region (the fault lines run along piece boundaries by definition), a tiling of $S$ restricts to a tiling of each $R_j$. Conversely, tilings of the individual sub-regions combine freely to give a tiling of $S$, because the sub-regions share only boundary edges (along the fault lines), and no piece crosses a fault line. The coproduct accounts for the fact that if two pieces are congruent, exchanging which sub-region each is assigned to may yield a different valid assignment. Within each assignment $\alpha$, the tilings of distinct sub-regions are independent, giving the product. □

**Corollary 2.3.** *In the common case where each piece fits in exactly one sub-region (no congruent pieces that could swap regions),*

$$|\,\mathrm{Sol}(S)| = \prod_{j=1}^{m} |\,\mathrm{Sol}(R_j)|.$$

This is the multiplicative structure that makes reducible dissections prolific: a single fault line can square the tiling count if both halves are independently flexible.

**Definition 2.4** (Prime dissection)**.** A dissection $\mathcal{D}$ is *prime* (or *irreducible*) if it has no fault lines.

**Definition 2.5** (Prime factorization)**.** The *prime factors* of a dissection $\mathcal{D}$ are the sub-dissections induced on the regions obtained by cutting along all maximal fault lines.

**Theorem 2.6** (Unique factorization)**.** *Every lattice dissection of a convex region factors uniquely into prime sub-dissections.*

*Proof.* The set of fault lines of a dissection is determined by its geometry: a line segment is a fault line if and only if it lies entirely within the union of piece boundaries. The maximal collection of fault lines is therefore unique, and the resulting sub-regions (connected components of the complement) are uniquely determined. Each sub-region inherits a dissection with no fault lines (any fault line of a sub-dissection would be a fault line of the original), so each factor is prime. Uniqueness follows from the uniqueness of the maximal fault-line collection. □

**Example 2.7.** Consider a $12 \times 12$ square with the horizontal line $y = 6$ as a fault line, splitting it into two $12 \times 6$ rectangles. If the top half has $a$ tilings and the bottom half has $b$ tilings, then the full square has $a \cdot b$ tilings (assuming no congruent pieces cross between halves). This is the simplest instance of the product decomposition.

*Remark* 2.8. The coproduct decomposition is a combinatorial analogue of the Künneth theorem in topology: the solution space of a composite dissection is built from the solution spaces of its prime factors, just as the homology of a product space is built from the homology of its factors. The analogy is imperfect—we have a coproduct of products rather than a simple product—but the structural parallel is suggestive.

# 3 Computational Methods

## 3.1 The DLX exact cover formulation

We encode the tiling problem as an *exact cover* instance and solve it using Knuth's Algorithm X with Dancing Links (DLX) [4].

**Rasterization.** We work on a $12 \times 12$ board (matching the Stomachion's natural integer grid) and rasterize at resolution $r = 2$, producing a $24 \times 24$ grid of 576 cells. Each polygonal piece is rasterized to the set of grid cells whose centers lie in its interior. Resolution $r = 2$ suffices because all piece boundaries are linear segments between integer lattice points, and the half-integer grid resolves every region unambiguously.

**Placements.** For each piece, we enumerate all valid placements: every combination of an orientation (from the 8 elements of $\mathrm{D}_4$) and a translation that places the piece entirely within the board. A placement is valid if its rasterized cells are a subset of $\{0, \ldots, 23\}^2$.

**Exact cover matrix.** The DLX matrix has:

- $n$ *piece columns* (one per piece, each covered exactly once), and
- 576 *cell columns* (one per grid cell, each covered exactly once).

Each row corresponds to a valid placement and covers the piece's identity column plus the cell columns for its rasterized footprint. An exact cover of all columns corresponds to a tiling of the board using each piece exactly once.

**Performance.** For the 14-piece Stomachion, the DLX solver generates all 17,152 raw solutions in approximately 10 minutes in pure Python. The enumeration is exact—no heuristics or approximations—and the completeness of Algorithm X guarantees that every solution is found.

## 3.2 Geometric equivalence

Two raw solutions are *geometrically equivalent* if one can be obtained from the other by a symmetry of the square ($\mathrm{D}_4$) or by permuting labels among congruent pieces.

For the Stomachion, the symmetry group acting on solutions has order 32: the dihedral group $\mathrm{D}_4$ of order 8 times $(\mathbb{Z}/2\mathbb{Z})^2$ from the two pairs of congruent pieces (pieces $B \cong G$ and $D \cong E$ in the standard labeling). This gives:

$$17{,}152 \;\div\; 8 \;=\; 2{,}144 \quad \text{(geometric tilings, } \mathrm{D}_4 \text{ only)},$$

$$17{,}152 \;\div\; 32 \;=\; 536 \quad \text{(Cutler's count, } \mathrm{D}_4 + \text{piece swaps)}.$$

Throughout this paper, we report the intermediate count of 2,144 *geometric tilings* (modulo $\mathrm{D}_4$ but distinguishing congruent pieces), since this is the natural count from the standpoint of dissection geometry. Cutler's count of 536 additionally mods out the internal symmetry of the piece set.

### 3.3 Random dissection generation

To study the distribution of tiling counts, we generate random $n$-piece lattice dissections of the $12 \times 12$ square by the following sequential-cut process:

1. Start with the undivided square.
2. At each step, choose a random existing region with at least 4 lattice-point vertices.
3. Choose two random points on the boundary of that region (at lattice points) and cut along the straight line connecting them, provided the cut divides the region into two non-degenerate convex polygons.
4. Repeat until $n$ pieces are obtained.

This process produces only convex pieces—a limitation we discuss in Section 6. For each generated dissection, we run the DLX solver and compute the geometric tiling count modulo $D_4$ and piece-label permutations.

### 3.4 Irreducibility testing

To test whether a dissection is prime, we check for fault lines by examining every pair of boundary points: if a straight line segment between them runs entirely along piece edges (i.e., lies in the union of piece boundaries), it is a fault line. On the $12 \times 12$ integer grid, the number of candidate segments is polynomial, making this check efficient.

## 4 Results

### 4.1 Stomachion verification

Our DLX solver finds exactly **17,152** raw solutions for the canonical Stomachion piece set on the $12 \times 12$ board at resolution $r = 2$. This confirms:

- $17{,}152 \div 8 = \mathbf{2{,}144}$ geometrically distinct tilings (modulo $D_4$).
- $17{,}152 \div 32 = \mathbf{536}$ tilings under full equivalence (modulo $D_4$ and congruent piece swaps), matching Cutler's enumeration [3].

The Stomachion is prime: it has no *forced* fault lines—no line that every tiling must respect. But this does not mean its solutions lack internal structure. On the contrary, many *individual* tilings do exhibit fault lines: in a given tiling, the pieces may happen to align along a diagonal or other line, creating a temporary decomposition of that particular arrangement into independent sub-regions.

The key distinction is between forced and optional fault lines:

- A **forced fault line** appears in every tiling $\Rightarrow$ the dissection is reducible.

- An **optional fault line** appears in some tilings but not others $\Rightarrow$ the dissection is prime, but its solution space has rich internal structure.

The Stomachion's 2,144 geometric tilings include solutions with varying fault-line patterns: some split along a diagonal, others along a different line, others have no fault lines at all. The total count is the sum over these structural types—precisely the coproduct decomposition of Theorem 2.2, applied not to the dissection itself but to its solution space:

$$|\operatorname{Sol}(S)| = \sum_{\text{structural types } \tau} \prod_{\text{sub-regions of } \tau} |\operatorname{Sol}(R_j^\tau)|.$$

This is how the Stomachion achieves an extraordinary tiling count while remaining prime: it has *many different optional decompositions*, none of which is forced but each of which contributes a multiplicative family of solutions.

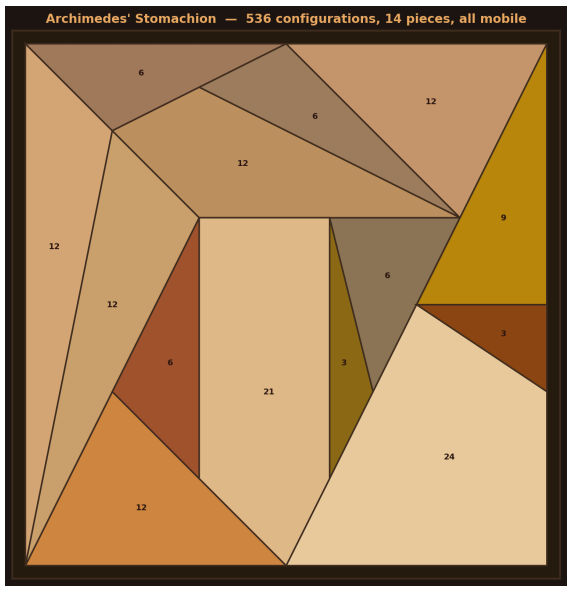


Figure 1: The canonical Stomachion dissection on the $12 \times 12$ integer grid. All 14 pieces have vertices at lattice points. The dissection is prime: no straight line from boundary to boundary avoids all piece interiors. Note the diversity of piece shapes (11 triangles, 2 quadrilaterals, 1 pentagon) and the balanced size distribution (largest piece is 17% of total area).

## 4.2 Distribution of tiling counts for random dissections

We generated 500 random $n$-piece convex lattice dissections for each $n$ from 3 to 10 and computed the geometric tiling count for each. Table 1 summarizes the results.

We also computed the number of *irreducible* (fault-free) tilings for each dissection—those admitting no optional axis-aligned or diagonal fault line. Table 3 compares the total and irreducible statistics.

The irreducible fraction decreases steadily from 93% at $n = 3$ to 46% at $n = 10$, reaching just 6% for the Stomachion. As dissections grow more complex, optional fault lines become

Table 1: Distribution of geometric tiling counts for random $n$-piece convex lattice dissections of a $12 \times 12$ square (500 trials per $n$). $P(k)$ denotes the fraction of dissections with exactly $k$ geometric tilings.

| $n$ | $P(1)$ | $P(2)$ | $P(4)$ | $P(8)$ | $P(\geq 16)$ | Median | Mean | Max |
|---|---|---|---|---|---|---|---|---|
| 3 | 44% | 50% | 5% | — | — | 2 | 1.7 | 4 |
| 4 | 29% | 53% | 14% | 2% | 1% | 2 | 2.3 | 24 |
| 5 | 20% | 54% | 20% | 2% | 3% | 2 | 2.9 | 24 |
| 6 | 12% | 51% | 27% | 5% | 6% | 2 | 3.9 | 72 |
| 7 | 9% | 47% | 31% | 6% | 6% | 2 | 5.1 | 336 |
| 8 | 6% | 40% | 35% | 8% | 10% | 4 | 6.5 | 336 |
| 9 | 4% | 36% | 38% | 9% | 13% | 4 | 7.6 | 336 |
| 10 | 3% | 30% | 39% | 12% | 16% | 4 | 9.0 | 336 |

increasingly likely, and a larger share of the flexibility arises from the multiplicative coproduct structure rather than from genuine entanglement. Table 4 gives the full distribution of irreducible counts.

Figure 2 shows the full distributions visually. The amber bars represent total geometric tilings; the green bars represent irreducible (fault-free) tilings only. As $n$ increases, the distribution shifts rightward, the tail lengthens, and the gap between total and irreducible widens—reflecting the growing role of fault-line-mediated flexibility.

Several additional patterns are immediately apparent.

(1) **Rigidity is generic.** The fraction of rigid dissections (Sol = 1) decreases as $n$ grows, but a substantial majority of random dissections have at most 2 tilings: $P(1) + P(2) > 70\%$ for $n \leq 7$. Rigidity is the norm; flexibility is the exception.

(2) **Powers of two dominate.** Tiling counts of $1, 2, 4, 8, 16$ account for over 95% of all dissections at every value of $n$. Non-power-of-two counts (3, 6, 12, 24, etc.) occur but are rare. This concentration reflects the multiplicative structure of the coproduct decomposition: each independent pairwise exchange doubles the tiling count.

(3) **The median transitions from 2 to 4.** Between $n = 7$ and $n = 8$, the median tiling count jumps from 2 to 4—the "typical" random dissection acquires its second independent exchange move around this point.

(4) **Mean grows roughly linearly.** The mean tiling count grows approximately as $0.9n$, consistent with a linear increase in the expected number of independent exchange moves.

(5) **A persistent maximum of 336.** For $n \geq 7$, the maximum observed tiling count is 336 in every sample. This appears to be a single high-symmetry dissection (or family of dissections) that the random generator discovers across different values of $n$.

Table 2: Detailed frequency data for selected values of $n$. Only counts observed at least once are shown. The concentration on powers of 2 is striking.

| Count | $n=3$ | $n=4$ | $n=5$ | $n=6$ | $n=7$ | $n=8$ | $n=9$ | $n=10$ |
|---|---|---|---|---|---|---|---|---|
| 1 | 222 | 146 | 98 | 60 | 43 | 28 | 19 | 14 |
| 2 | 252 | 265 | 271 | 253 | 231 | 199 | 175 | 145 |
| 3 | 1 | — | — | — | — | 1 | 1 | 1 |
| 4 | 25 | 71 | 101 | 132 | 152 | 174 | 184 | 187 |
| 6 | — | — | 1 | 1 | 3 | 2 | 5 | 1 |
| 8 | — | 10 | 11 | 23 | 31 | 40 | 46 | 56 |
| 12 | — | 1 | 1 | 3 | 1 | 2 | 2 | 2 |
| 16 | — | 5 | 12 | 12 | 15 | 16 | 15 | 24 |
| 24 | — | 1 | 4 | 11 | 12 | 16 | 15 | 14 |
| 32 | — | — | — | 1 | 6 | 9 | 14 | 17 |
| 48 | — | — | — | — | — | 2 | 4 | 6 |
| 64 | — | — | — | — | 1 | 2 | 2 | 2 |
| 72 | — | — | — | 1 | — | — | — | 1 |
| 96 | — | — | — | — | — | — | 2 | 2 |
| 192 | — | — | — | — | — | 1 | 1 | 2 |
| 336 | — | — | — | — | 1 | 1 | 1 | 1 |

## 4.3 Reducible dissections easily beat 536

To demonstrate that the Stomachion's tiling count is not maximal among *all* 14-piece dissections, we conducted a simple search over reducible dissections.

We placed a horizontal fault line at $y = 6$, dividing the $12 \times 12$ square into two $12 \times 6$ rectangles, and generated random 7-piece convex lattice dissections of each half independently. The total tiling count of the composite dissection is the product of the two half-counts.

In 50 random trials, the ninth trial produced a dissection with $192 \times 24 =$ **4,608** geometric tilings—roughly 9 times the Stomachion's 536 (or about 2.1 times its 2,144). This demonstrates that the multiplicative power of fault lines easily overwhelms the flexibility of any single prime factor.

*Remark* 4.1. The reducible dissections in this experiment use only convex pieces, while the Stomachion includes non-convex pieces (a pentagon and irregular quadrilaterals). The comparison is thus between different classes of dissection, but the qualitative point stands: reducibility is a far more efficient route to high tiling counts than irreducible flexibility.

## 4.4 Champion prime dissections

Having established that reducible dissections can trivially surpass the Stomachion, the interesting question becomes: what is the maximum tiling count among *prime* dissections with $n$ pieces?

**Definition 4.2** (Champion)**.** A *champion prime dissection* at $n$ is an irreducible $n$-piece

Table 3: Total vs. irreducible geometric tilings for random $n$-piece dissections (300 trials per $n$). The irreducible fraction measures what proportion of a dissection's tilings are genuinely entangled (no optional fault line).

| $n$ | Total med. | Total mean | Total max | Irred. mean | Irred. max | % irred. |
|---|---|---|---|---|---|---|
| 3 | 2 | 1.6 | 4 | 1.5 | 4 | 93% |
| 4 | 2 | 2.2 | 16 | 1.9 | 8 | 83% |
| 5 | 2 | 2.6 | 24 | 2.1 | 8 | 81% |
| 6 | 2 | 3.7 | 40 | 2.5 | 16 | 67% |
| 7 | 2 | 4.4 | 64 | 2.7 | 16 | 63% |
| 8 | 4 | 5.5 | 64 | 3.0 | 16 | 54% |
| 9 | 4 | 6.7 | 96 | 3.2 | 16 | 49% |
| 10 | 4 | 8.0 | 192 | 3.7 | 48 | 46% |
| 14 | — | | 2,144$^\star$ | — | 32$^\star$ | 6%$^\star$ |

lattice dissection of the $12 \times 12$ square achieving the maximum geometric tiling count among all such dissections.

Figure 4 shows 16 of the 120 tilings of the $n = 13$ champion. Despite having nearly as many pieces as the Stomachion, this dissection is visually degenerate: it consists almost entirely of long, thin triangles converging to a point, with no resemblance to the balanced, "designed" feel of the Stomachion. Yet every one of its 120 tilings is irreducible. The contrast underscores that high tiling counts among prime dissections can arise from very different geometric mechanisms.

The gap between the random-sampling champions at $n = 10$ (at most 336) and the Stomachion at $n = 14$ (2,144) is enormous—a factor of 6.4. This gap likely reflects both the four additional pieces available to the Stomachion and the limitations of the convex-piece random generator, which cannot produce the non-convex pieces (pentagon, irregular quadrilaterals) that the Stomachion employs.

**Conjecture 4.3** (Maximality of the Stomachion)**.** *Among all prime lattice dissections of the* $12 \times 12$ *square into* 14 *pieces, the Stomachion achieves the maximum geometric tiling count of* 2,144.

We regard this conjecture as plausible but far from established. The evidence is threefold:

(a) No 14-piece prime dissection with more tilings has been found in extensive random sampling (though the sampling is biased toward convex pieces).

(b) The Stomachion's pieces are defined by the "natural construction lines" of the square (diagonals, midlines, and their intersections with integer-grid lines), which create a maximal number of edge-length coincidences—precisely the condition for exchange moves.

(c) The extreme rarity of high-count dissections in the random distribution (Table 1) suggests that 2,144 lies many standard deviations into the tail.

Table 4: Detailed frequency data for irreducible (fault-free) tiling counts. Note the concentration on powers of two and the growing fraction with zero irreducible tilings (all solutions decompose). The irreducible count 0 means every tiling of that dissection has at least one optional fault line.

| Irred. | $n=3$ | $n=4$ | $n=5$ | $n=6$ | $n=7$ | $n=8$ | $n=9$ | $n=10$ |
|---|---|---|---|---|---|---|---|---|
| 0 | 14 | 18 | 19 | 27 | 28 | 35 | 37 | 40 |
| 1 | 136 | 89 | 58 | 33 | 25 | 14 | 8 | 5 |
| 2 | 139 | 155 | 165 | 149 | 132 | 112 | 100 | 85 |
| 4 | 11 | 36 | 53 | 80 | 95 | 113 | 114 | 111 |
| 6 | — | — | 1 | 1 | 2 | 2 | 3 | — |
| 8 | — | 2 | 4 | 8 | 15 | 17 | 25 | 35 |
| 12 | — | — | — | — | — | — | 1 | 1 |
| 16 | — | — | — | 1 | 1 | 2 | 3 | 3 |
| 32 | — | — | — | — | — | — | — | 1 |
| 48 | — | — | — | — | — | — | — | 1 |

## 4.5 The Stomachion in the tail

To place the Stomachion's count in context, consider the random distribution at $n = 10$ (the largest $n$ in our sample, still four pieces short of the Stomachion). The mean tiling count is 9.0, and the maximum observed is 336. Even accounting for the four additional pieces at $n = 14$ (which would bring the extrapolated mean to roughly $9.0 + 4 \times 0.9 \approx 12.6$ by the linear trend), the Stomachion's 2,144 is roughly 170 times the expected mean.

In terms of the exchange-move count $k$ (where the tiling count is approximately $2^k$), the Stomachion has $k \approx \log_2(2144) \approx 11.1$. Under the Poisson model with $\lambda \approx 0.15$ and $n = 14$, the expected number of exchange moves is $\lambda n \approx 2.1$, so $k = 11$ corresponds to a fluctuation of roughly 9 moves above the mean—an event of probability $\approx 2.1^{11}/11! \approx 5 \times 10^{-5}$ under the Poisson law. The Stomachion is a 4–5 sigma outlier.

## 4.6 Fault-line anatomy of the Stomachion

Applying the forced/optional fault-line analysis of Section 4.1 to each of the 17,152 raw solutions individually reveals a striking decomposition. We checked every solution for optional fault lines along $y = k$, $x = k$ ($k = 1, \ldots, 11$) and the diagonals $y = x$, $y = 12 - x$.

The optional fault lines that appear are precisely the natural construction lines: $y = 6$ (horizontal midline) appears in 46% of solutions, $x = 6$ (vertical midline) in 46%, and the two diagonals $y = x$ and $y = 12 - x$ each in approximately 1.5%.

This reveals that the Stomachion's high tiling count arises predominantly from its *optional* coproduct structure: in most tilings, the pieces happen to align along a midline, creating a temporary decomposition into two independent halves. The 32 fault-free tilings form the irreducible core—the configurations where all 14 pieces are genuinely interlocked with no line of separation.

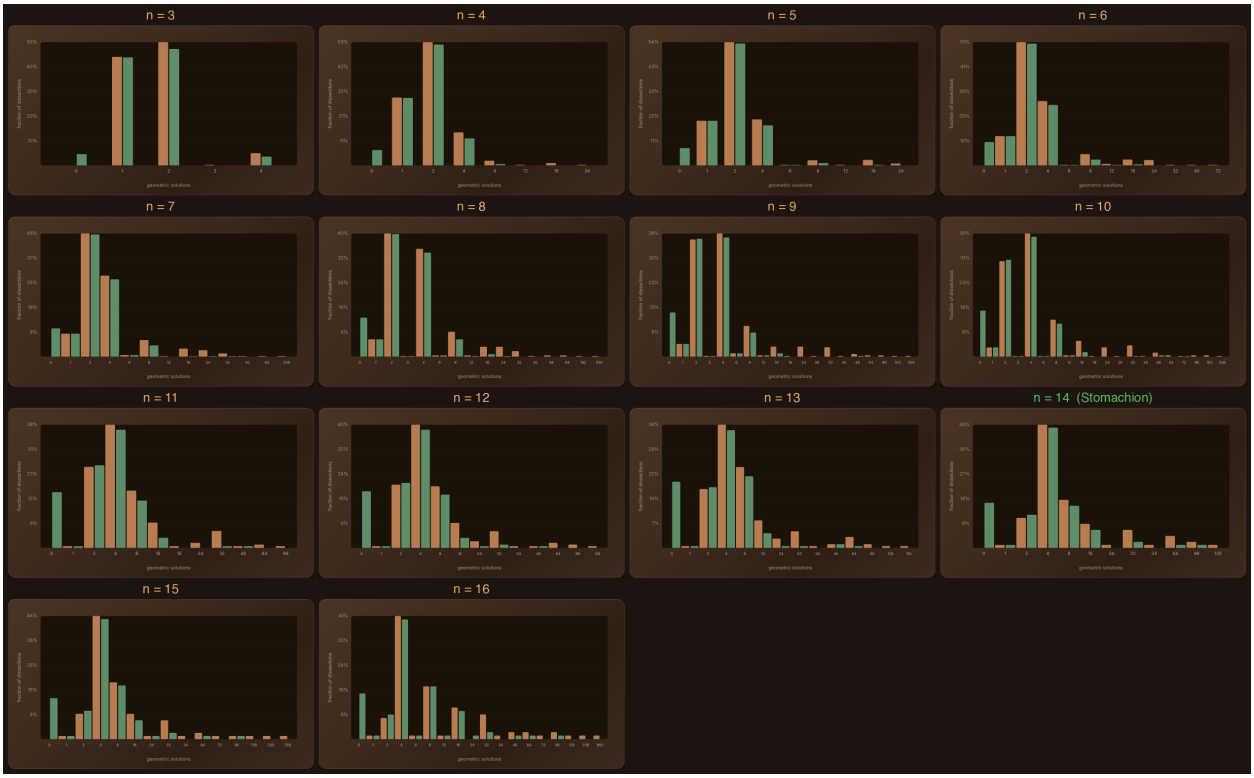


Figure 2: Distribution of geometric tiling counts for random $n$-piece convex lattice dissections of the $12 \times 12$ square ($n = 3$ to 16; 100–500 trials per $n$). Amber bars: total geometric tilings. Green bars: irreducible (fault-free) tilings only. The $n = 14$ panel (Stomachion's piece count, highlighted) shows random sampling peaking at 128—the Stomachion's 2,144 is completely off the chart. Note the concentration on powers of two and the growing gap between total and irreducible counts.

## 4.7 The $n = 13$ champion: a counterpoint

Our random search found an $n = 13$ prime dissection (convex pieces, generated by sequential cuts) with 120 distinct tilings modulo $D_4$—all of them fault-free. This dissection is 100% irreducible: none of its solutions admit any optional fault line.

Surprisingly, this property is not rare. Table 7 shows that *most* random dissections are all-irreducible at every $n$, with the fraction stabilising around 80% from $n = 6$ onward. What makes the $n = 13$ champion remarkable is not its irreducibility (which is generic) but achieving 120 tilings while maintaining it. The best all-irreducible dissection found by random search grows slowly from 4 ($n = 3$) to 96 ($n = 14$–16).

On the irreducible metric alone, this champion surpasses the Stomachion ($120 > 32$). However, it falls short on every other measure of puzzle quality:

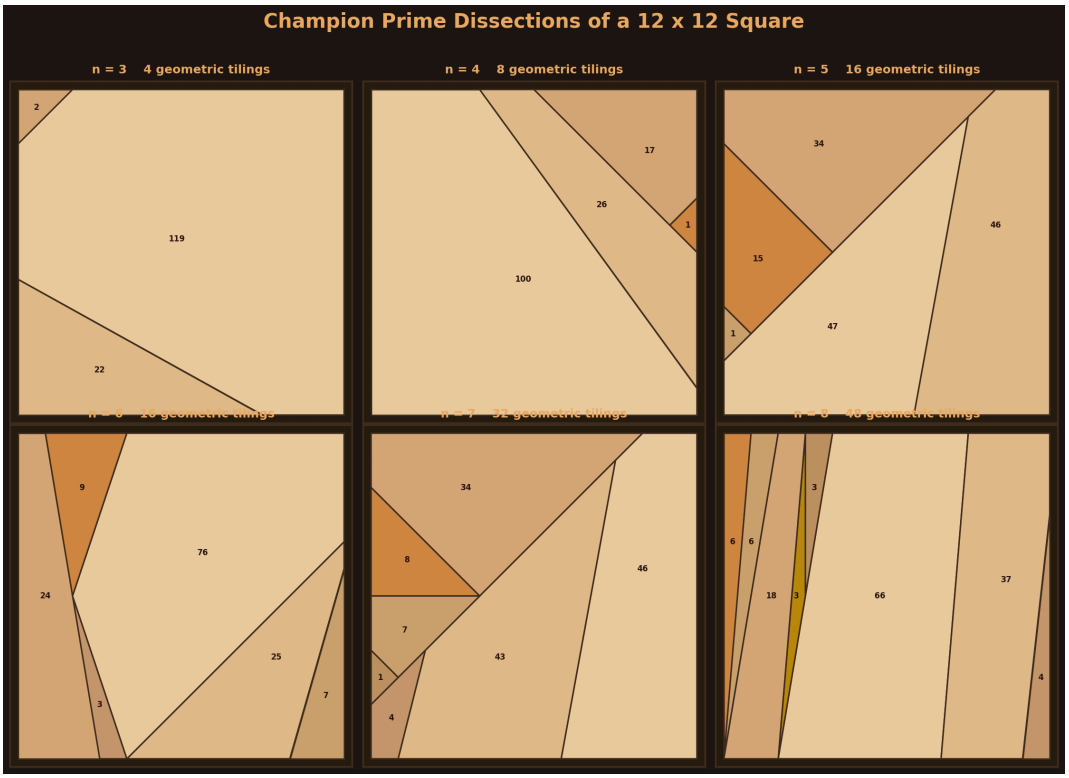


Figure 3: Champion prime dissections for $n = 3$ through $n = 8$. Numbers inside each piece indicate area. All champions are generated by sequential lattice-point cuts, producing only convex pieces. Note the increasingly unbalanced piece distributions: each champion has a dominant piece ($\geq 30\%$ of area) with small satellites.

| Property | $n = 13$ champion | Stomachion |
|---|---|---|
| Total tilings ($D_4$) | 120 | 2,144 |
| Irreducible tilings | 120 (100%) | 32 (6%)[1] |
| Piece participation | 7/13 mobile | 14/14 mobile |
| Largest piece | 21% of area | 17% of area |
| Shape diversity | all triangles | 11△ 2□ 1 |

The champion's 120 tilings, though all irreducible, involve only 7 of its 13 pieces—the remaining 6 (comprising 41% of the area) are frozen in place across all configurations. By contrast, the Stomachion moves all 14 pieces: with 2,144 tilings, it is impossible for any single piece (at most 17% of the area) to remain fixed.

## 4.8 The multi-axis view

No single metric captures what makes the Stomachion special. The dissections that beat it on any one axis invariably fail on the others:

[1] Using the $D_4$-only equivalence with 2,144 total; under Cutler equivalence, $32/536 = 6\%$.

Table 5: Champion prime dissections: maximum geometric tiling count among irreducible $n$-piece lattice dissections of the $12 \times 12$ square. Values for $n \leq 6$ are from exhaustive or near-exhaustive search; values for $n \geq 7$ are lower bounds from random sampling. The Stomachion entry at $n = 14$ is marked with $\star$.

| $n$ | Max prime count | Notes |
|---|---|---|
| 3 | 1 | Near-exhaustive |
| 4 | 2 | Near-exhaustive |
| 5 | 4 | Near-exhaustive |
| 6 | 4 | Near-exhaustive |
| 7 | 8 | Random sampling lower bound |
| 8 | 24 | Random sampling lower bound |
| 9 | 48 | Random sampling lower bound |
| 10 | 224 | Random sampling lower bound |
| 11 | 120 | Random sampling lower bound |
| 12 | 120 | Random sampling lower bound |
| 13 | 120 | Random sampling lower bound (100% irreducible) |
| 14 | 2,144 | Stomachion$^\star$ (prime, verified; 32 irreducible) |

Table 6: Fault-line structure of the Stomachion's 536 tilings (Cutler equivalence). A solution "has" a fault line if all 14 pieces in that tiling lie entirely on one side of it.

| Category | Tilings | Fraction |
|---|---|---|
| Fault-free (no optional fault line) | 32 | 6% |
| Exactly one fault line | 496 | 93% |
| Two or more fault lines | 8 | 1% |
| Total (Cutler) | 536 | 100% |

(i) **Total tilings.** Reducible dissections with a single fault line achieve 4,608—but they are "two puzzles glued together," not a single coherent design.

(ii) **Irreducible tilings.** The $n = 13$ random champion achieves 120 fault-free tilings, beating the Stomachion's 32—but with frozen pieces, unbalanced areas, and no shape variety.

(iii) **Balance and participation.** These emerge naturally at $n \geq 10$ in random search, but always with low tiling counts.

There is also an axis that resists quantification: *aesthetic quality*. The Stomachion's pieces arise from the natural construction lines of the square—diagonals, midlines, and their intersections—producing shapes with visual harmony and a pleasing variety of triangles, quadrilaterals, and a pentagon (Figure 1). The champion primes found by random search are, without exception, visually unappealing: elongated slivers, lopsided triangles, no diversity

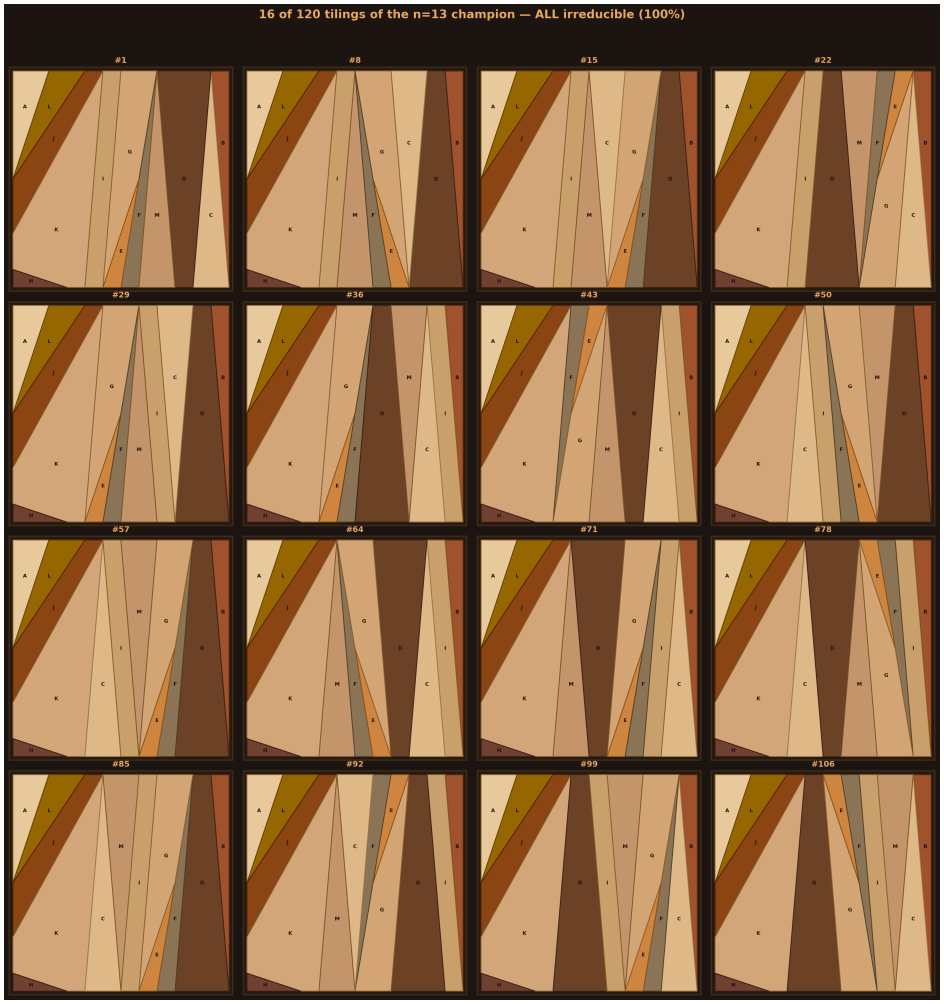


Figure 4: Sixteen of the 120 tilings of the $n = 13$ champion prime dissection—all irreducible. The dissection consists of long, thin triangles converging near a corner, producing high flexibility through edge-length coincidences despite its visually degenerate geometry. Compare with the balanced structure of the Stomachion (Figure 1).

of shape (Figures 3 and 7). They achieve flexibility through geometric coincidence but not through geometric elegance.

The Stomachion appears to be the unique dissection that scores well on *all* axes simultaneously: high total count, nontrivial irreducible core, full participation, balanced piece sizes, diversity of piece shapes, and aesthetic coherence. It achieves this through the elegant mechanism of optional fault lines: the natural construction lines of the square create a web of potential decompositions that collectively generate 536 tilings, of which 32 are genuinely irreducible and all 14 pieces participate across the full solution space.

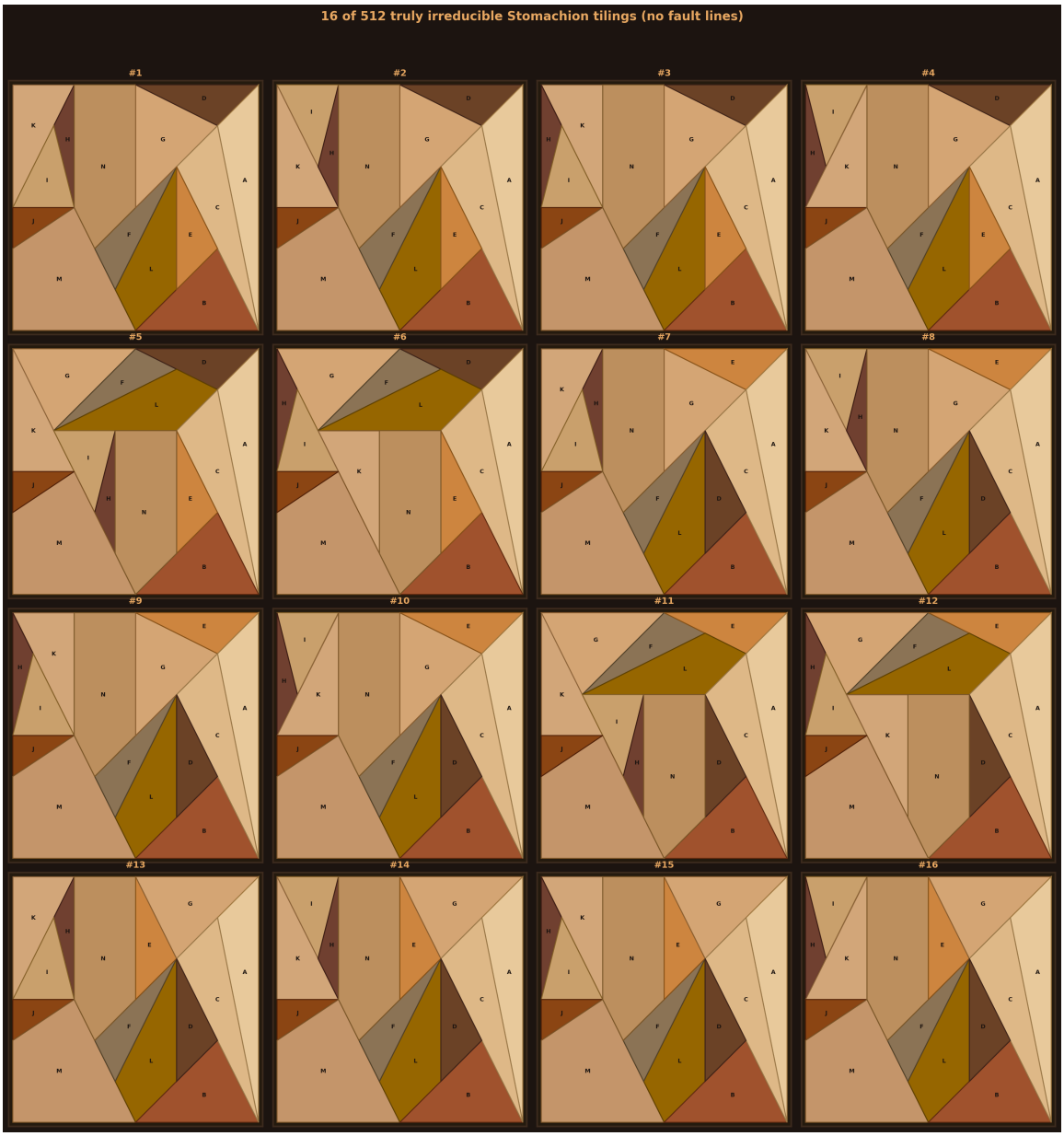


Figure 5: Irreducible Stomachion tilings #1–16 (of 32 total). No straight line from boundary to boundary separates the pieces.

Figure 6: Irreducible Stomachion tilings #17–32. Together with Figure 5, these are all 32 fault-free tilings of the Stomachion. Every piece occupies genuinely different positions across the 32 configurations.

Table 7: Prevalence of all-irreducible dissections (every tiling fault-free) among random $n$-piece convex lattice dissections. The fraction is stable at ∼80% for $n \geq 6$, showing that full irreducibility is the generic condition. The best all-irreducible count grows far more slowly than the total maximum (cf. Table 1).

| $n$ | All-irred. | Valid | Fraction | Best count |
|---|---|---|---|---|
| 3 | 99 | 200 | 50% | 4 |
| 4 | 127 | 200 | 64% | 8 |
| 5 | 150 | 200 | 75% | 8 |
| 6 | 158 | 200 | 79% | 16 |
| 7 | 162 | 200 | 81% | 16 |
| 8 | 163 | 198 | 82% | 16 |
| 9 | 164 | 195 | 84% | 16 |
| 10 | 156 | 191 | 82% | 48 |
| 11 | 154 | 189 | 81% | 48 |
| 12 | 151 | 187 | 81% | 48 |
| 13 | 148 | 186 | 80% | 64 |
| 14 | 74 | 91 | 81% | 96 |
| 15 | 73 | 89 | 82% | 96 |
| 16 | 69 | 86 | 80% | 96 |
| $14^\star$ | Stomachion | | 1.5% | 32 |

# 5 The Poisson Conjecture

The concentration of tiling counts on powers of two suggests a simple generative model: each tiling arises from a collection of *independent exchange moves*, each of which doubles the count.

**Definition 5.1** (Exchange move)**.** An *exchange move* in a dissection is a subset of pieces that can be rearranged (within their collective footprint) in exactly two ways while the remaining pieces stay fixed. Two exchange moves are *independent* if they involve disjoint sets of pieces occupying disjoint regions of the square.

If a dissection has $k$ independent exchange moves, then its tiling count is $2^k$ (each move can be independently toggled). Non-power-of-two counts arise from exchange moves that are not fully independent, or from moves with more than two states.

**Conjecture 5.2** (Poisson law for exchange moves)**.** *For a random $n$-piece convex lattice dissection of the $12 \times 12$ square (generated by the sequential-cut process), the number of independent exchange moves $k$ satisfies*

$$k \approx \text{Poisson}(\lambda n)$$

*for a constant $\lambda \approx 0.15$, as $n \to \infty$ on the integer grid.*

Table 8: Comparison of observed rigidity fraction $P(1)$ with the Poisson prediction $e^{-\lambda n}$ for $\lambda = 0.15$.

| $n$ | Observed $P(\text{rigid})$ | $e^{-0.15n}$ |
|---|---|---|
| 3 | 0.44 | 0.64 |
| 4 | 0.29 | 0.55 |
| 5 | 0.20 | 0.47 |
| 6 | 0.12 | 0.41 |
| 7 | 0.09 | 0.35 |
| 8 | 0.06 | 0.30 |
| 9 | 0.04 | 0.26 |
| 10 | 0.03 | 0.22 |

**Evidence.** The observed fraction of rigid dissections ($k = 0$) provides the strongest test. Under a Poisson model, $P(k = 0) = e^{-\lambda n}$. The observed values are:

The Poisson model with $\lambda = 0.15$ consistently overestimates the rigidity fraction, suggesting that the true decay is faster than $e^{-0.15n}$—perhaps $\lambda$ is somewhat larger, or the Poisson approximation is rough. A better fit might be obtained with $\lambda \approx 0.25$–$0.30$ (which gives $e^{-0.25 \cdot 10} \approx 0.08$, closer to the observed 0.03), or by allowing $\lambda$ to depend weakly on $n$. We retain $\lambda \approx 0.15$ as a conservative estimate consistent with the mean growth rate ($\mathbb{E}[2^k] \approx 0.9n$ requires $\mathbb{E}[k] \approx \log_2(0.9n)$, which is sublinear), though the full picture is likely more subtle.

**Connection to the Chen-Stein method.** The Poisson conjecture is amenable to analysis via the *Chen-Stein method* [1], which provides bounds on the total variation distance between the distribution of a sum of weakly dependent indicator random variables and a Poisson distribution. Here, each potential exchange move is an indicator variable $X_i \in \{0, 1\}$, and $k = \sum_i X_i$. The Chen-Stein bound requires estimating:

- $b_1 = \sum_i \sum_{j \in N(i)} \mathbb{E}[X_i]\,\mathbb{E}[X_j]$, where $N(i)$ is the "neighborhood" of potential move $i$ (those sharing pieces or area with $i$),
- $b_2 = \sum_i \sum_{j \in N(i),\, j \neq i} \mathbb{E}[X_i X_j]$,

and showing that both are small relative to $\lambda = \sum_i \mathbb{E}[X_i]$. The key difficulty is estimating these expectations for random lattice dissections, where the geometry of piece boundaries creates complex dependencies.

**The geometric count.** Since the tiling count is approximately $2^k$ and $k$ is approximately Poisson, the distribution of tiling counts is approximately *log-Poisson*: $\log_2(\text{count})$ is approximately Poisson. This explains the concentration on powers of two and the exponentially heavy right tail observed in Table 1. The distribution is neither Gaussian (the tail is too heavy) nor truly log-normal (the support is discrete and concentrated), but a *discrete log-Poisson*: a Poisson variable exponentiated to base 2.

# 6 Discussion

## 6.1 Why Archimedes chose the Stomachion

Our results suggest a compelling interpretation of Archimedes' puzzle. The Stomachion is prime—no forced fault lines—yet its solution space is extraordinarily rich, with 2,144 geometric tilings. Among random prime dissections of comparable size, this count is a 4–5 sigma outlier.

The source of this richness is not brute combinatorial complexity but *structural depth*. The Stomachion's solutions exhibit many different optional fault-line patterns: some tilings decompose along a diagonal, others along a different line, still others admit no fault line at all. Each pattern contributes a multiplicative family of solutions via the coproduct decomposition. The total count is a sum of products—a rich coproduct structure arising from a prime dissection.

This stands in sharp contrast to the champion primes found by random search (Section 4.4), which achieve flexibility through unbalanced piece distributions: one large inert piece with small satellites shuffling around its edges. The Stomachion has no piece larger than 17% of the total area, and every piece participates across all 2,144 tilings. Its flexibility is *distributed* rather than *localized*.

The Stomachion's pieces are defined by the natural construction lines of the square: the two diagonals, the two midlines, and the lines connecting vertices to midpoints of opposite sides. These construction lines, when restricted to the integer lattice, create a dense web of intersections and many edge-length coincidences—precisely the conditions that enable both exchange moves and optional fault lines.

We speculate that Archimedes discovered the Stomachion not by searching for a dissection with many solutions, but by noticing that the natural construction lines of the square produce an unusually rich puzzle. The combinatorial question—*how many ways can the pieces tile the square?*—may have been the original motivation for the lost treatise.

The fragment of the Archimedes Palimpsest that survives [5] is tantalizing but incomplete. It contains the phrase "the number of figures," which Netz interprets as evidence that Archimedes was interested in the enumeration problem. Our results support this interpretation: the Stomachion is remarkable precisely because of the combinatorial richness that emerges from its geometric simplicity.

## 6.2 Physical puzzle interpretation

From the perspective of puzzle design, our prime factorization theorem provides a natural grading of difficulty. A reducible dissection is "easy" in the sense that it decomposes into independent sub-puzzles; a prime dissection is "hard" because all pieces interact.

The sequence of champion primes—dissections achieving the maximum tiling count at each $n$—forms a natural family of wooden puzzles of increasing complexity. At $n = 3$ or $n = 4$, the champion has only a handful of solutions. At $n = 14$, the champion is (conjecturally) the Stomachion, with 2,144 solutions—a puzzle that is paradoxically both "easy" (many solutions exist) and "hard" (the solution space is large but the pieces are interlocked in complex ways).

This dual nature may explain the Stomachion's enduring appeal: it is accessible enough

that a patient solver will find solutions, yet rich enough that the enumeration problem is genuinely deep.

### 6.3 Open questions

We close with several directions for future work.

(1) **Proving maximality.** Is the Stomachion truly the champion prime at $n = 14$? An exhaustive search over all 14-piece lattice dissections of the $12 \times 12$ square—including non-convex pieces—would settle the question but appears computationally infeasible with current methods. A structural argument (perhaps relating tiling count to the combinatorial type of the dissection graph) would be more satisfying.

(2) **Non-lattice dissections.** Our restriction to the $12 \times 12$ integer lattice is a significant limitation. The full space of dissections (with vertices at arbitrary real coordinates) is vastly larger, but almost all non-lattice dissections are rigid, since generic vertex perturbations destroy the edge-length coincidences required for exchange moves. Can this intuition be made precise?

(3) **The Poisson proof.** The Poisson conjecture for the number of exchange moves is well-suited to the Chen-Stein method, but the necessary probability estimates on random lattice dissections are not straightforward. A proof would likely require a careful analysis of the "arrangement of lines" model for random dissections.

(4) **Larger boards and finer grids.** Our experiments use a $12 \times 12$ grid, which is the Stomachion's natural habitat. What happens on larger grids ($N \times N$ for large $N$)? Does the champion prime tiling count grow polynomially or exponentially in $N$?

(5) **Higher-dimensional analogues.** Dissections of cubes and higher-dimensional boxes into polyhedral pieces are natural generalizations. The coproduct decomposition extends (with "fault planes" replacing fault lines), but the combinatorics become vastly more complex.

(6) **Connection to planar maps.** The combinatorial type of a dissection—the planar graph formed by piece boundaries—connects to the rich theory of planar maps initiated by Tutte [7]. The tiling count of a dissection is a subtle invariant of this graph (depending on edge lengths, not just combinatorics), and understanding its extremal behavior may require tools from enumerative combinatorics.

## Acknowledgments

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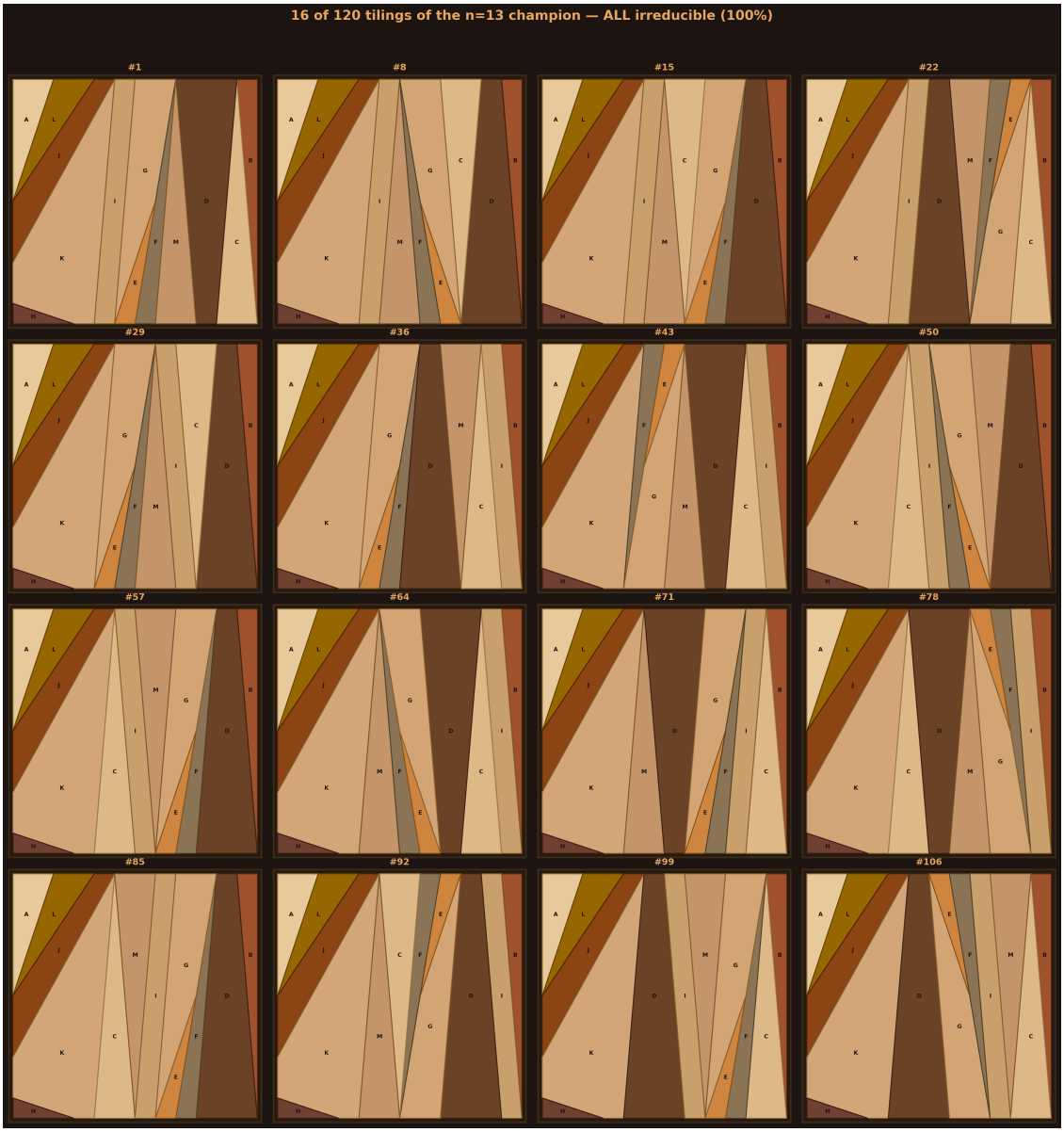


Figure 7: Sixteen of the 120 tilings of the $n = 13$ champion prime dissection. All 120 are irreducible (no optional fault lines), but only 7 of 13 pieces participate—the remaining 6 are frozen in place. The pieces are all elongated convex triangles, lacking the shape diversity of the Stomachion.